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Exp: 14 Code using RAW socket to implement packet sniffing

Aim:

To create a program using RAW sockets to capture and display network packets passing through the network interface.

Algorithm:

- Run the program, and create a Raw packet using socket (AF_PACKET, SOCKET_RAW, ..)
- Bind the socket to the required network interface, then continously receive packets from the network.
- Extract the ethernet header from each packet and identify protocol type, such as IPv4, ARP, or IPv6
- For IPv4 packets, extract the source and the destination IP addresses.
- Display the packet information, Repeat until the user terminates the prog

Code :

```
import socket
import struct
def get_protocol(protocol_num):
    protocols = {
        1: "ICMP",
        6: "TCP",
        17: "UDP"
    }
    return protocols.get(protocol_num, "Other")
def packet_sniffer():
    try:
        sniffer = socket.socket(
            socket.AF_INET,
            socket.SOCK_RAW,
            socket.IPPROTO_IP
        )
        host = socket.gethostbyname(socket.gethostname())
        sniffer.bind((host, 0))
        sniffer.setsockopt(
            socket.IPPROTO_IP,
            socket.IP_HDRINCL,
            1
        )
        print("Packet Sniffer Started...")
        print("Listening for network packets...\n")
        while True:
            packet, address = sniffer.recvfrom(65565)
            ip_header = packet[0:20]
            iph = struct.unpack("!BBHHHBBH4s4s", ip_header
            version_ihl = iph[0]
            print("Listening for network packets...\n")
            while True:
                packet, address = sniffer.recvfrom(65565)
                ip_header = packet[0:20]
                iph = struct.unpack("!BBHHHBBH4s4s", ip_header)
                version_ihl = iph[0]
                version = version_ihl >> 4
                ihl = (version_ihl & 15) * 4
                ttl = iph[5]
                protocol = iph[6]
                source_ip = socket.inet_ntoa(iph[8])
                destination_ip = socket.inet_ntoa(iph[9])
                print("----- Packet Captured -----")
                print("IP Version       :", version)
                print("Header Length    :", ihl)
                print("Source IP        :", source_ip)
                print("Destination IP   :", destination_ip)
                print("Protocol         :", get_protocol(protocol))
                print("TTL              :", ttl)
                print("Packet Length    :", len(packet), "bytes")
                print("---------------------------\n")
    except PermissionError:
        print("Error: Run the program with administrator/root privileges.")
    except KeyboardInterrupt:
        print("\nPacket Sniffing Stopped.")
    except OSError as error:
        print("Socket Error:", error)
    finally:
        if 'sniffer' in locals():
            sniffer.close()
```

Output:

```
Packet Sniffer Started...
Listening for network packets...

----- Packet Captured -----
IP Version        : 4
Header Length     : 20
Source IP         : 192.168.1.5
Destination IP    : 142.250.183.14
Protocol          : TCP
TTL               : 64
Packet Length     : 60 bytes
---------------------------

----- Packet Captured -----
IP Version        : 4
Header Length     : 20
Source IP         : 192.168.1.5
Destination IP    : 8.8.8.8
```

Result:

The code to implement RAW sockets to implement packet sniffing has been completed.